Analysis for BELLS

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Introduction

Bell's palsy is an acute peripheral facial nerve paralysis that can result in incomplete recovery and long-term morbidity. Corticosteroids and antiviral agents have been proposed as treatments, but their effectiveness has remained uncertain.

Study Objective

To determine whether prednisolone or acyclovir used early in the course of Bell's palsy improves the chances of recovery

Data and methods

Data source

The data used in this analysis are from a randomized, double-blind, placebo-controlled clinical trial that investigated the effectiveness of prednisolone and acyclovir in patients with Bell's palsy. The dataset contains individual-level participant data including demographic characteristics, baseline disease severity, treatment allocation, and recovery outcomes.

Study design

Participants were randomly assigned using a 2x2 factorial design to either of four groups, to receive prednisolone, acyclovir, both treatments, or placebo. The primary outcome was facial nerve recovery assessed after follow-up.

Data preparations

```
bells <- read.csv("bells.csv", check.names = FALSE)
```

The dataset was imported into R using the 'read.csv()' function and stored as an object named bells.This object contains participant level information used in the subsequent analysis.

Initial data inspection

```r
dim(bells)
```

```
## [1] 494  12
```

I used the dim() function to determine the number of observations/rows(494) and variables/columns(12)

494 observations are noted which is fewer than the 496 mentioned in the trial publication, as having outcomes

```r
str(bells)
```

```
## 'data.frame':    494 obs. of  12 variables:
##  $ Patient ID                                          : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Sex                                                 : chr  "Female" "Female" "Female" "Female" .
##  $ Age                                                 : int  77 61 46 46 42 50 37 28 41 36 ...
##  $ Baseline Score on House-Brackmann scale             : int  6 6 4 3 3 4 4 4 4 4 ...
##  $ Time between onset of symptoms and start of treatment: chr  "Within 24 hr" "Within 24 hr" ">24 to
##  $ Treatment Group                                     : chr  "Prednisolone-Placebo" "Prednisolone-
Placebo" "Prednisolone-Placebo" "Prednisolone-Placebo" ...
##  $ Received Prednisolone                               : chr  "Yes" "Yes" "Yes" "Yes" ...
##  $ Received Acyclovir                                  : chr  "No" "No" "No" "No" ...
##  $ 3-Month Score on House-Brackmann scale              : int  2 1 1 1 1 1 1 1 1 1 ...
##  $ Full Recovery in 3 Months                           : chr  "No" "Yes" "Yes" "Yes" ...
##  $ 9-Month Score on House-Brackmann scale              : int  2 1 1 1 1 1 1 1 1 1 ...
##  $ Full Recovery in 9 Months                           : chr  "No" "Yes" "Yes" "Yes" ...
```

I used the str() function to determine the structure of the data and identify variables that needed further processing

Data processing

```r
cat_var <- c("Sex", "Time between onset of symptoms and start of treatment", "Treatment Group", "Receiv

bells[cat_var] <- lapply(bells[cat_var], factor)

summary(bells)
```

```
##    Patient ID        Sex           Age
##  Min.   :  1.0   Female:238   Min.   :16.00
##  1st Qu.:124.2   Male  :256   1st Qu.:34.00
##  Median :247.5                Median :44.00
##  Mean   :247.5                Mean   :44.87
##  3rd Qu.:370.8                3rd Qu.:55.00
##  Max.   :494.0                Max.   :90.00
##  Baseline Score on House-Brackmann scale
##  Min.   :2.00
##  1st Qu.:3.00
##  Median :4.00
##  Mean   :3.68
##  3rd Qu.:4.00
##  Max.   :6.00
```

```
##  Time between onset of symptoms and start of treatment
##  >24 to  48 hr:205
##  >48 to  72 hr: 39
##  Unknown      :  1
##  Within 24 hr :249
##
##
##                Treatment Group Received Prednisolone Received Acyclovir
##  Acyclovir-Placebo     :122    No :244               No :245
##  Acyclovir-Prednisolone:127    Yes:250               Yes:249
##  Placebo-Placebo       :122
##  Prednisolone-Placebo  :123
##
##
##  3-Month Score on House-Brackmann scale Full Recovery in 3 Months
##  Min.   :1.00                           No :136
##  1st Qu.:1.00                           Yes:358
##  Median :1.00
##  Mean   :1.34
##  3rd Qu.:2.00
##  Max.   :4.00
##  9-Month Score on House-Brackmann scale Full Recovery in 9 Months
##  Min.   :1.000                          No : 53
##  1st Qu.:1.000                          Yes:441
##  Median :1.000
##  Mean   :1.144
##  3rd Qu.:1.000
##  Max.   :4.000
```

Within the dataset, several variables required explicit conversion to factor type to ensure correct handling as categorical variables in subsequent analyses. A character vector of variable names was therefore created using the c() function, and these variables were simultaneously converted to factors using lapply(). The summary() function was then used to confirm that the conversions were successful.

```
colSums(is.na(bells))
```

```
##                                            Patient ID
##                                                     0
##                                                   Sex
##                                                     0
##                                                   Age
##                                                     0
##               Baseline Score on House-Brackmann scale
##                                                     0
## Time between onset of symptoms and start of treatment
##                                                     0
##                                         Treatment Group
##                                                     0
##                                   Received Prednisolone
##                                                     0
##                                      Received Acyclovir
##                                                     0
##                3-Month Score on House-Brackmann scale
```

```
##                                                 0
##                             Full Recovery in 3 Months
##                                                 0
##               9-Month Score on House-Brackmann scale
##                                                 0
##                             Full Recovery in 9 Months
##                                                 0
```

```
summary(bells$Age)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   16.00   34.00   44.00   44.87   55.00   90.00
```

```
summary(bells$`Baseline Score on House-Brackmann scale`)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    2.00    3.00    4.00    3.68    4.00    6.00
```

```
summary(bells$`3-Month Score on House-Brackmann scale`)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1.00    1.00    1.00    1.34    2.00    4.00
```

```
summary(bells$`9-Month Score on House-Brackmann scale`)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   1.000   1.000   1.000   1.144   1.000   4.000
```

```
table(bells$`Full Recovery in 3 Months`, bells$`Full Recovery in 9 Months`)
```

```
##
##        No Yes
##   No   53  83
##   Yes   0 358
```

Data completeness was assessed by quantifying missing values across all variables using. Validity of age was assessed using range checks to confirm values were non-negative and within a biologically plausible range.Validity of House–Brackmann scores was assessed at baseline, 3 months, and 9 months. Baseline scores ranged from 2 to 6, while follow-up scores ranged from 1 to 6, consistent with the expected scale limits

The logical consistency of the data was also assessed.Its not plausible for a participant to have recovered(Yes) at 3 months(rows) and not recovered(No) at 9 months (columns)

Descriptive analysis

```
summary(bells$Age)
```

Baseline

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   16.00   34.00   44.00   44.87   55.00   90.00
```

The median age of the participants was 44.0 with an Interquartile range of 34-55

```
prop.table(table(bells$Sex))
```

```
##
##    Female      Male
## 0.4817814 0.5182186
```

Of the 494 participants, 48.2% were female and 51.8% were male

```
addmargins(table(bells$`Treatment Group`))
```

```
##
##      Acyclovir-Placebo Acyclovir-Prednisolone        Placebo-Placebo
##                    122                     127                    122
##   Prednisolone-Placebo                     Sum
##                    123                     494
```

Using the 2x2 factorial design, participants were randomized to the different treatments group and evenly distributed as shown in the table above showing that randomization was successful

```
addmargins(table(bells$`Full Recovery in 3 Months`))
```

Outcomes

```
##
##  No Yes Sum
## 136 358 494
```

```
prop.table(table(bells$`Full Recovery in 3 Months`)) * 100
```

```
##
##       No      Yes
## 27.53036 72.46964
```

After 3 months; 72.5% of the participants had recovered

```
addmargins(table(bells$`Full Recovery in 9 Months`))
```

```
##
##  No Yes Sum
##  53 441 494
```

Characteristic	Acyclovir–Placebo N = 122[1]	Acyclovir–Prednisolone N
Age	44 (34, 54)	45 (35, 54)
Sex		
Female	56 (46%)	62 (49%)
Male	66 (54%)	65 (51%)
Baseline Score on House–Brackmann scale		
2	17 (14%)	24 (19%)
3	28 (23%)	37 (29%)
4	48 (39%)	37 (29%)
5	20 (16%)	23 (18%)
6	9 (7.4%)	6 (4.7%)
Time between onset of symptoms and start of treatment		
>24 to 48 hr	41 (34%)	53 (42%)
>48 to 72 hr	12 (9.8%)	12 (9.4%)
Unknown	1 (0.8%)	0 (0%)
Within 24 hr	68 (56%)	62 (49%)

[1]Median (Q1, Q3); n (%)

```
prop.table(table(bells$`Full Recovery in 9 Months`)) * 100
```

```
##
##       No      Yes
## 10.72874 89.27126
```

After 9 months; 89.3% of participants had recovered

Baseline characteristics

The baseline characteristics of the participants will now be assessed for each treatment group to determine if randomization was successful

```
theme_gtsummary_compact()
```

```
## Setting theme "Compact"
```

```
bells %>%
  select(`Treatment Group`, Age, Sex, `Baseline Score on House-Brackmann scale`, `Time between onset of
  tbl_summary(`Treatment Group`)
```

The baseline characteristics of the 494 participants were evenly distributed across the 4 treatment groups as shown in the table above. Median Age(approximately 44 years old) across the four groups was similar. The male to female ratio across the four groups was roughly similar. The median Baseline score on House-Brackmann scale was 4 for all the four groups. Additionally, the time between onset of symptoms and start of treatment was equally distributed amongst the four groups. This proves that randomisation was succesful

Exploratory Analysis

Test for Interaction We shall first test the data for amongst the study groups using Logistic Regression model;

```r
#Build Logistic regression model
interaction_model <- glm(`Full Recovery in 9 Months` ~ `Received Prednisolone`* `Received Acyclovir` + A
tbl_regression(
  interaction_model,
  exponentiate = TRUE,
  label = list(
    `Received Prednisolone` = "Received Prednisolone",
    `Received Acyclovir`    = "Received Acyclovir"
  )
)
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
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## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

Logistic regression was run to test for interaction between prednisolone and Acyclovir. It was adjusted for by Age, Sex, Time between onset and treatment and Baseline severity. A p value of 0.6 confirms that no interaction occured if patients took both prednisolone and acyclovir. This means interaction between the two drugs was not significant

Characteristic	**OR**	**95% CI**	**p-value**
Received Prednisolone			
No	—	—	
Yes	2.89	1.16, 7.94	0.028
Received Acyclovir			
No	—	—	
Yes	1.05	0.49, 2.26	0.9
Age	0.96	0.94, 0.98	<0.001
Sex			
Female	—	—	
Male	0.63	0.33, 1.16	0.14
Baseline Score on House–Brackmann scale	0.71	0.54, 0.94	0.016
Time between onset of symptoms and start of treatment			
>24 to 48 hr	—	—	
>48 to 72 hr	0.59	0.18, 2.32	0.4
Unknown	0.00		>0.9
Within 24 hr	0.65	0.33, 1.24	0.2
Received Prednisolone * Received Acyclovir			
Yes * Yes	0.72	0.19, 2.62	0.6

Abbreviations: CI = Confidence Interval, OR = Odds Ratio